

Exercise Sheet 13

Discrete Mathematics by Hongfei Fu, 2023.11.23

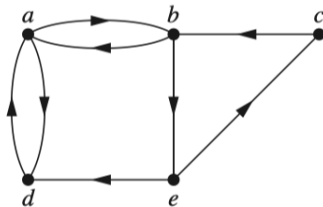
Name: _____ Student ID: _____ Class: _____

1. Does each of these lists of vertices form a path in the following graph? Which paths are simple? Which are circuits? What are the lengths of those that are paths?

a) a, b, e, c, b

b) a, d, a, d, a

d) a, b, e, c, b, d, a



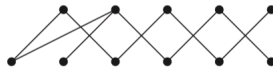
Answer Area:

2. How many connected components does each of the graphs in Exercises 3-5 have? For each graph find each of its connected components.

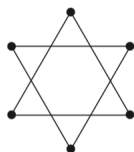
3.



4.

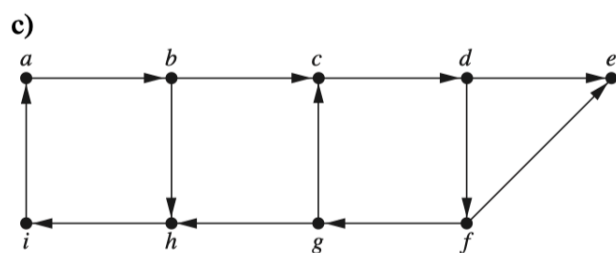
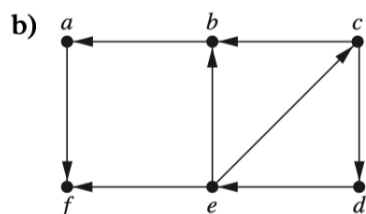
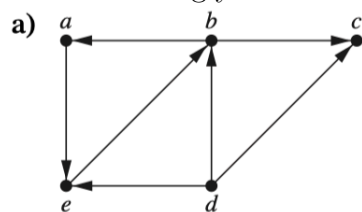


5.



Answer Area:

3. Find the strongly connected components of each of these graphs.



Answer Area:

4. (2pt) Show that every connected simple graph with n vertices has at least $n - 1$ edges.

Answer Area:

5. Show that a vertex c in the connected simple graph G is a cut vertex if and only if there are vertices u and v , both different from c , such that every path between u and v passes through c . **Answer Area:**